Grade 6 Accelerated
Unit 1
Lesson 1 Practice

Name Answer Key_________
Date_____________________
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Find the greatest common factor (GCF) of each pair of numbers.

1. 36 and 54

$36: 1, 2, 3, 4, 6, 9, 12, 18, 36$
$54: 1, 2, 3, 6, 9, 18, 27, 54$

GCF $= 18$

2. 17 and 71

$17: 1, 17$
$71: 1, 71$

GCF $= 1$

3. 84 and 315

$84: 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84$
$315: 1, 3, 5, 7, 9, 15, 21, 35, 45, 63, 105, 315$

GCF $= 21$

4. 374 and 682

$374: 1, 2, 11, 17, 22, 34, 187, 374$
$682: 1, 2, 11, 22, 31, 62, 341, 682$

GCF $= 22$

5. Jehzaria made hair bows for the girls track season. She made 30 blue bows and 45 orange bows. She wants to make gift bags for the team that have the same amount of blue and orange bows in each bag. Jehzaria wants to make as many gift bags as possible without having any bows left over. What is the largest number of gift bags that she can make?

$30: 1, 2, 3, 5, 6, 10, 15, 30$
$45: 1, 3, 5, 9, 15, 45$

She can make 15 bags.

Rewrite each fraction in simplest form using the GCF of the numerator and denominator.

6. $\frac{42}{60}$

$\frac{42}{60} = \frac{\cancel{6}\times 7}{\cancel{6}\times 10} = \frac{7}{10}$

7. $\frac{80}{272}$

$\frac{80}{272} = \frac{5\times\cancel{16}}{17\times\cancel{16}} = \frac{5}{17}$

8. $\frac{23}{138}$

$\frac{23}{138} = \frac{1\times\cancel{23}}{6\times\cancel{23}} = \frac{1}{6}$

9. $\frac{558}{620}$

$\frac{558}{620} = \frac{9\times\cancel{62}}{10\times\cancel{62}} = \frac{9}{10}$

10. Camille was looking at the fraction $\frac{525}{700}$ and says that the fraction rewritten in simplest form is $\frac{105}{140}$ because she knows that $525 \div 5 = 105$ and $700 \div 5 = 140$. Explain to Camille why she is incorrect.

It can be simplified even more.

$$\frac{105}{140} = \frac{3 \times \cancel{35}}{4 \times \cancel{35}} = \frac{3}{4}$$

Grade 6 Accelerated
Unit 1
Lesson 2 Practice

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Find the least common multiple (LCM) of each pair of numbers.

1. 12 and 15

$12: 12, 24, 36, 48, 60$
$15: 15, 30, 45, 60$

LCM $= 60$

2. 7 and 11

$7: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77$
$11: 11, 22, 33, 44, 55, 66, 77$

LCM $= 77$

3. 35 and 5

$5: 5, 10, 15, 20, 25, 30, 35$
$35: 35$

LCM $= 35$

4. 24 and 16

$16: 16, 32, 48$
$24: 24, 48$

LCM $= 48$

5. 10 and 25

$10: 10, 20, 30, 40, 50$
$25: 25, 50$

LCM $= 50$

6. 9 and 21

$9: 9, 18, 27, 36, 45, 54, 63$
$21: 21, 42, 63$

LCM $= 63$

7. The volleyball team has a game every 6 days, while the basketball team has a game every 4 days. If both teams had a game today, in how many days will they next have games on the same day?

$4: 4, 8, 12$
$6: 6, 12$

They will both have games in 12 days.

8. A school librarian is preparing for the book fair, and the books have just arrived. She has one pile of boxes where each box is 20 inches tall and another pile of boxes where each box is 24 inches tall. If the librarian stacks boxes so that both piles are the same height, what is the minimum height of the piles?

$20: 20, 40, 60, 80, 100, 120$
$24: 24, 48, 72, 96, 120$

The minimum height is 120 inches.

9. A grocery store is having a promotion where every 15^{th} customer receives a $\$50$ gift card and every 25^{th} customer receives free groceries for a month. What number customer will be the first to receive both the gift card and the free groceries for a month?

$15: 15, 30, 45, 60, 75$
$25: 25, 50, 75$

The 75^{th} customer will receive both.

10. At the Earlington Heights station on the Miami-Dade Metrorail, an orange line train arrives every 14 minutes, and a green line train arrives every 8 minutes. If a train from each line arrives at $7{:}00$ AM, at what time will trains from both lines next arrive simultaneously?

$8: 8, 16, 24, 32, 40, 48, 56$
$14: 14, 28, 42, 56$

Trains from both lines will arrive simultaneously in 56 minutes when it will be $7{:}56$ AM.

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Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
1. 41×8	$8(40+1)$ $= 320 + 8$ $= 328$	40 + 1 8: 320 | 8	328
2. 6×89	$6(80+9)$ $= 480 + 54$ $= 534$	80 + 9 6: 480 | 54	534
3. 96×3	$3(90+6)$ $= 270 + 18$ $= 288$	90 + 6 3: 270 | 18	288

For each expression, generate an equivalent numeric expression using the distributive property and then find the value.

4. 9×42

$9(40+2)$
$= 9 \times 40 + 9 \times 2$
$= 360 + 18$
$= 378$

5. $28 \cdot 7$

$7(20+8)$
$= 7 \times 20 + 7 \times 8$
$= 140 + 56$
$= 196$

6. 5×64

$5(60+4)$
$= 5 \times 60 + 5 \times 4$
$= 300 + 20$
$= 320$

Rewrite an equivalent expression for each sum using the distributive property with the greatest common factor (GCF).

7. $90 + 75$

$15(6 + 5)$

8. $208 + 336$

$16(13 + 21)$

9. Generate two different equivalent expressions for the sum $340 + 220$ by rewriting it using the distributive property with two different common factors.

Answers may vary.

$2(170 + 110)$
$10(34 + 22)$

10. Jayden used the distributive property to write an expression equivalent to $48 + 72$. His expression was $4(8 + 18)$. Is Jayden correct? If yes, explain how he determined his expression. If not, explain Jayden's error and write a correct equivalent expression.

He is not correct because $4 \times 8 = 32$ not 48. It should be $4(12 + 18)$.

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Unit 1
Lesson 4 Practice

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Find each product using the area model.

1. $(18.3)(7.9) = 144.57$

	7	0.9
18	126	16.2
0.3	2.1	0.27

2. $(9.25)(12.7) = 117.475$

	12	0.7
9	108	6.3
0.25	3	0.175

3. Find the product 61.8×0.29 using partial products.

$$\begin{array}{r}
61.8 \\
\times\ 0.29 \\
\hline
\left.\begin{array}{r} 0.072 \\ 0.09 \\ 5.4 \end{array}\right] = 5.562 \\
\left.\begin{array}{r} 0.16 \\ 0.2 \\ 12.0 \end{array}\right] = 12.36 \\
\hline
17.922
\end{array}$$

Find the product $5.39 \cdot 4.64$ using the specified method.

4. Partial Products	5. Standard Algorithm
$\begin{array}{l} \ 5.39 \\ \underline{\times\ 4.64} \\ \left.\begin{array}{l} 0.0036 \\ 0.012 \\ 0.2 \end{array}\right] = 0.2156 \\ \left.\begin{array}{l} 0.054 \\ 0.18 \\ 3.0 \end{array}\right] = 3.234 \\ \left.\begin{array}{l} 0.36 \\ 1.2 \\ 20.0 \end{array}\right] = 21.56 \\ \hline 25.0096 \end{array}$	$\begin{array}{r} 5.39 \\ \times\ 4.64 \\ \hline 0.2156 \\ 3.234 \\ 21.56 \\ \hline 25.0096 \end{array}$

Find each product using any method.

6. $(82.3)(4.5) = 370.35$

	4	0.5
82	328	41
0.3	1.2	0.15

7. $27.54 \times 3.91 = 107.6814$

$$\begin{array}{r} 27.54 \\ \times\ 3.91 \\ \hline 0.2754 \\ 24.786 \\ 82.62 \\ \hline 107.6814 \end{array}$$

8. $8.185 \cdot 7.2 = 58.932$

$$\begin{array}{r} 8.185 \\ \times\ 7.2 \\ \hline 1.637 \\ 57.295 \\ \hline 58.932 \end{array}$$

9. $(3.08)(0.289) = 0.89012$

	3	0.08
0.289	0.867	0.02312

10. Horatio made a mistake when using the standard algorithm to find the product 18.35×6.2. Write Horatio a note explaining his mistake.

$$\begin{array}{cccccc} & & \mathbf{1} & \mathbf{8.} & \mathbf{3} & \mathbf{5} \\ & \times & & \mathbf{6.} & \mathbf{2} & \\ \hline & & 3 & 6. & 7 & 0 \\ + & 1 & 1 & 0. & 1 & 0 \\ \hline & \mathbf{1} & \mathbf{4} & \mathbf{6.} & \mathbf{8} & \mathbf{0} \end{array}$$

The decimal is misplaced. It should be 3.67 which would give a final product of 113.77.

Grade 6 Accelerated
Unit 1
Lesson 5 Practice

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Find each quotient. Show your work.

1. $48.64 \div 0.8 = 60.8$

$$\begin{array}{r} 60.8 \\ 08\overline{)486.4} \\ \underline{-48} \\ 6 \\ \underline{-0} \\ 64 \\ \underline{-64} \\ 0 \end{array}$$

2. $49.02 \div 3.8 = 12.9$

$$\begin{array}{r} 12.9 \\ 38\overline{)490.2} \\ \underline{-38} \\ 110 \\ \underline{-76} \\ 342 \\ \underline{-342} \\ 0 \end{array}$$

3. $217.74 \div 0.38 = 573$

$$\begin{array}{r} 573 \\ 38\overline{)21774} \\ \underline{-190} \\ 277 \\ \underline{-266} \\ 114 \\ \underline{-114} \\ 0 \end{array}$$

4. $48.861 \div 9.15 = 5.34$

$$\begin{array}{r} 5.34 \\ 915\overline{)4886.10} \\ \underline{-4575} \\ 3111 \\ \underline{-2745} \\ 3660 \\ \underline{-3660} \\ 0 \end{array}$$

5. $95.03 \div 11.18 = 8.5$

$$\begin{array}{r} 8.5 \\ 1118\overline{)9503.0} \\ \underline{-8944} \\ 5590 \\ \underline{-5590} \\ 0 \end{array}$$

6. $41.752 \div 61.4 = 0.68$

$$\begin{array}{r} 0.68 \\ 614\overline{)417.52} \\ \underline{-3684} \\ 4912 \\ \underline{-4912} \\ 0 \end{array}$$

7. Joel was finding the quotient $68.82 \div 3.7$, but he knows his answer doesn't make sense. His work is shown. Write a note to Joel explaining his mistake.

$$\begin{array}{r} 1.86 \\ 37\overline{)\,68.82} \\ -37 \\ \hline 318 \\ -296 \\ \hline 222 \\ -222 \\ \hline 000 \end{array}$$

When Joel multiplied the divisor by 10 to get 37, he did not multiply the dividend, 68.82, by 10.

An expression is shown. Use the expression to answer questions 8-10.

$$132.03 \div 8.1$$

8. Which of the expression is equivalent to $132.03 \div 8.1$? Select all that apply.

- ☒ $1320.3 \div 81$
- ☐ $13203 \div 81$
- ☐ $13.203 \div 81$
- ☒ $13203 \div 810$
- ☒ $13.203 \div 0.81$

9. Select one of your answer choices from question 8 and explain why it is equivalent to $132.03 \div 8.1$.

Answers may vary.

$1320.3 \div 81$ is equivalent to $132.03 \div 8.1$ because both the dividend and divisor have been multiplied by 10.

10. Find the quotient $132.03 \div 8.1$.

$$\begin{array}{r} 16.3 \\ 81\overline{)1320.3} \\ -81 \\ \hline 510 \\ -486 \\ \hline 243 \\ -243 \\ \hline 0 \end{array}$$

Grade 6 Accelerated
Unit 1
Lesson 6 Practice

Name Answer Key
Date
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Find each product using the area model.

1. $1\frac{3}{5} \times 2\frac{1}{4} = 2\frac{32}{20} = 3\frac{3}{5}$

	2	$\frac{1}{4}$
1	2	$\frac{1}{4} = \frac{5}{20}$
$\frac{3}{5}$	$\frac{6}{5} = \frac{24}{20}$	$\frac{3}{20}$

2. $4\frac{2}{3} \times 3\frac{1}{2} = 16\frac{1}{3}$

	3	$\frac{1}{2}$
4	12	2
$\frac{2}{3}$	$\frac{6}{3} = 2$	$\frac{2}{6} = \frac{1}{3}$

3. $5\frac{3}{4} \times 8\frac{1}{5} = 47\frac{3}{20}$

	8	$\frac{1}{5}$
5	40	1
$\frac{3}{4}$	6	$\frac{3}{20}$

Find each product. Rewrite each product in simplest form. Show your work.

4. $\frac{3}{16} \times \frac{4}{5} = \frac{3}{20}$

$= \frac{12}{80} = \frac{3 \times \cancel{4}}{20 \times \cancel{4}} = \frac{3}{20}$

5. $\frac{7}{11} \times 3\frac{2}{3} = 2\frac{1}{3}$

$= \frac{7}{11} \times \frac{11}{3} = \frac{77}{33} = \frac{7 \times \cancel{11}}{3 \times \cancel{11}} = \frac{7}{3} = 2\frac{1}{3}$

6. $4\frac{2}{5} \times 4\frac{4}{9} = 19\frac{5}{9}$

$= \frac{22}{5} \times \frac{40}{9} = \frac{880}{45} = \frac{176 \times \cancel{5}}{9 \times \cancel{5}} = \frac{176}{9} = 19\frac{5}{9}$

7. $\frac{37}{100} \times 3\frac{1}{2} = 2\frac{59}{200}$

$= \frac{37}{100} \times \frac{7}{2} = \frac{259}{200} = 2\frac{59}{200}$

8. $6\frac{5}{12} \times \frac{4}{5} = 5\frac{2}{15}$

$= \frac{77}{12} \times \frac{4}{5} = \frac{308}{60} = \frac{77 \times \cancel{4}}{15 \times \cancel{4}} = \frac{77}{15} = 5\frac{2}{15}$

9. $8\frac{7}{20} \times 4\frac{9}{10} = 40\frac{183}{200}$

$= \frac{167}{20} \times \frac{49}{10} = \frac{8183}{200} = 40\frac{183}{200}$

10. Andre was working on the multiplication expression $3\frac{1}{2} \times 4\frac{3}{5}$. His work is shown. Is Andre correct or incorrect? If he is correct, explain the steps that he took. If he is incorrect, explain his mistake(s).

$$3\frac{1}{2} \times 4\frac{3}{5}$$
$$\frac{7}{2} \times \frac{23}{5}$$
$$\frac{7 \times 23}{2 \times 5}$$
$$\frac{161}{10} \text{ or } 16\frac{1}{10}$$

He is correct.
1. He changed the mixed numbers to fractions.
2. He multiplied the numerators to get the numerator of the product and then did the same with the denominators.
3. The answer was then put into a mixed number in simplest form.

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Lesson 7 Practice

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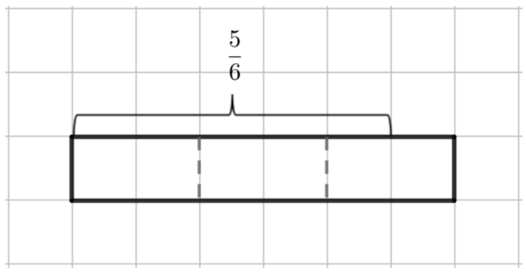
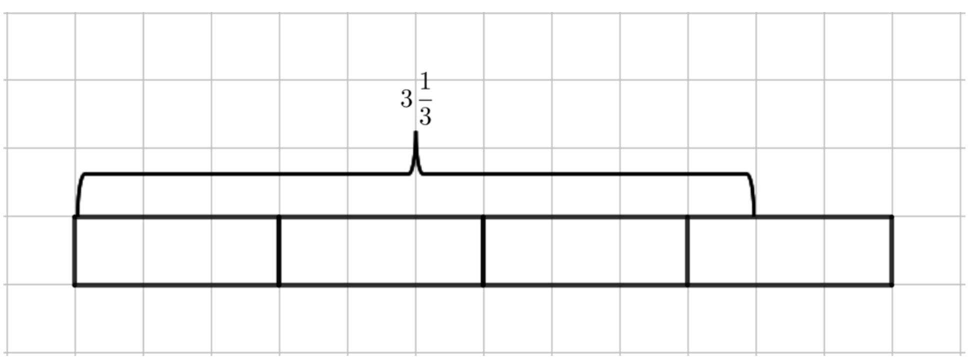
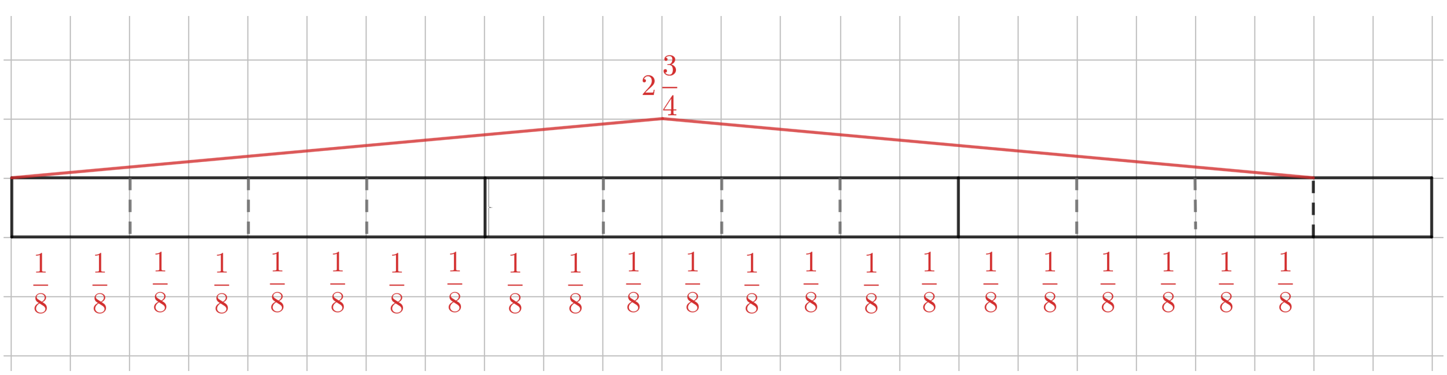
Complete the table by finding each quotient. Use the tape diagram shown.

Tape Diagram	Division Expression	Quotient
1. 	$\frac{5}{6} \div \frac{1}{3}$	$\frac{5}{2} = 2\frac{1}{2}$
2. 	$3\frac{1}{3} \div \frac{2}{3}$	5

3. Draw a tape diagram to find the quotient of $2\frac{3}{4} \div \frac{1}{8}$.



$2\frac{3}{4} \div \frac{1}{8} = 22$

Find each quotient. Simplify when possible.

4. $\frac{3}{8} \div \frac{5}{6} = \frac{9}{20}$

$= \frac{3}{8} \times \frac{6}{5} = \frac{18}{40} = \frac{9}{20}$

5. $\frac{5}{6} \div \frac{3}{8} = 2\frac{2}{9}$

$= \frac{5}{6} \times \frac{8}{3} = \frac{40}{18} = \frac{20}{9} = 2\frac{2}{9}$

6. $\frac{6}{11} \div 9 = \frac{2}{33}$

$= \frac{6}{11} \times \frac{1}{9} = \frac{6}{99} = \frac{2}{33}$

7. $5 \div \frac{15}{34} = 11\frac{1}{3}$

$= \frac{5}{1} \times \frac{34}{15} = \frac{170}{15} = \frac{34}{3} = 11\frac{1}{3}$

8. $5\frac{2}{9} \div \frac{3}{10} = 17\frac{11}{27}$

$= \frac{47}{9} \times \frac{10}{3} = \frac{470}{27} = 17\frac{11}{27}$

9. $3\frac{7}{8} \div 1\frac{1}{2} = 2\frac{7}{12}$

$= \frac{31}{8} \times \frac{2}{3} = \frac{62}{24} = \frac{31}{12} = 2\frac{7}{12}$

10. Mireya used the standard algorithm to divide $2\frac{7}{10} \div 1\frac{3}{8}$. Her work is shown. Is Mireya correct? If she is correct, explain the steps that she took. If she is incorrect, explain her mistake and fix it.

Step 1: $2\frac{7}{10} \div 1\frac{3}{8}$

Step 2: $\frac{27}{10} \div \frac{11}{8}$

Step 3: $\frac{10}{27} \times \frac{8}{11}$

Step 4: $\frac{80}{297}$

Mireya is incorrect because she used the reciprocal of both fractions when she only should have used the reciprocal of the second fraction. The correct answer is:

$\frac{27}{10} \times \frac{8}{11} = \frac{216}{110} = \frac{108}{55} = 1\frac{53}{55}$